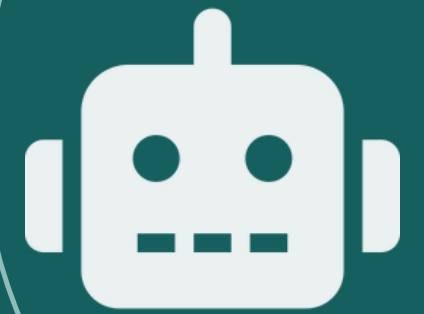




AI JOB MARKET

A Data-Driven Analysis of Global AI Hiring Trends, Salaries & Skills

PREPARED BY

Noran Hatem • Aya Ayman • Dalia Khaled

Ziad Ashraf • Kirolos Ibrahim • Arwa Ahmed

Supervised by: Eng. Dina Mohsen

Python • SQL • Power BI

What We'll Walk Through

01

Project Introduction

Goals and problem framing

02

Dataset Overview

Scope, fields and coverage

03

Python Workflow

Exploration, cleaning, validation, visualization

04

SQL Analysis

Techniques and business questions

05

Power BI Dashboards

Objectives, audiences, KPIs & DAX

06

Dashboard Insights

Key findings & business conclusions

07

Final Deliverables

GitHub repository structure

Understanding the Global AI Job Market

The rapid growth of Artificial Intelligence has reshaped the global job market — creating new roles, shifting required skill sets, and driving significant salary premiums. This project analyzes a global AI job-postings dataset end-to-end, moving from raw data to a clean analytical dataset, structured SQL analysis, and interactive Power BI dashboards.

PROJECT GOAL



Explore, clean and analyze global AI job postings to uncover patterns in roles, salaries, required skills and market demand — then translate those patterns into decision-ready dashboards for job seekers, recruiters and analysts.



Job Roles & Categories



Salary & Compensation



Skills & Requirements



Demand & Market Trends

Global AI Jobs Dataset

2000

Job Postings

37

Data Fields

25

Distinct AI Job Titles

12

Job Categories

KEY DATA FIELDS

Job Details

job_title, job_category, experience_level, years_of_experience, education_required

Compensation

annual_salary_usd, salary_min/max_usd, ai_salary_premium_pct, salary_tier

Location & Setup

city, country, remote_work, company_size, industry

Skills & Demand

required_skills, skill_count, demand_score, demand_growth_yoy_pct, is_llm_role

From Raw Data to Analysis-Ready Dataset



Notebook: [ai_jobs_global_processed.ipynb](#)

[open in google colab](#)

1

Data Exploration

Loaded the dataset in Pandas and inspected shape, columns, data types, and initial structure to understand the raw data.

2

Data Cleaning

Removed whitespace, standardized missing-like values, checked duplicates, and converted date columns into proper formats.

3

Data Validation

Validated numeric ranges, categorical consistency, salary logic, binary indicator fields and detected outliers using IQR & Z-score.

4

Data Visualization & Feature Engineering

Engineered features such as skill_count, salary_range_midpoint and posting_date, then profiled distributions to prepare the dataset for SQL and Power BI.



Turning Data Into Business Answers

Four SQL scripts were written against the cleaned dataset — one per dashboard — combining aggregate functions, GROUP BY, JOIN logic, CASE statements, subqueries and window-style analysis to answer 28 targeted business questions.

SQL TECHNIQUES USED

Aggregate Functions
(COUNT, AVG, MIN, MAX)

GROUP BY & ORDER BY
ranking analysis

JOINS & CASE
conditional logic

Subqueries
nested comparisons

STRING_SPLIT +
CROSS APPLY for skills

TOP N
filtering

FOUR SQL DASHBOARDS OF ANALYSIS

Dashboard 1

AI Job Market Overview

4 KPIs · 7 questions

Dashboard 2

Salary & Compensation Analysis

4 KPIs · 7 questions

Dashboard 3

AI Skills & Career Requirements

4 KPIs · 7 questions

Dashboard 4

AI Job Demand & Market Trends

4 KPIs · 7 questions

AI Job Market Overview

OBJECTIVE

Provide an overall view of the AI job market and identify major trends in job roles, categories, locations, industries and experience levels.

TARGET AUDIENCE

Job Seekers • Career Analysts • Recruiters

Main KPIs: Total Job Postings • Total Job Roles • Total Job Categories • Average Annual Salary

1,500

Total Job Postings

25

Total Job Roles

12

Total Job Categories

\$194,892

Average Annual Salary

Top 5 Job Titles by Postings



Top 5 Countries by Job Postings





AI Job Market Overview

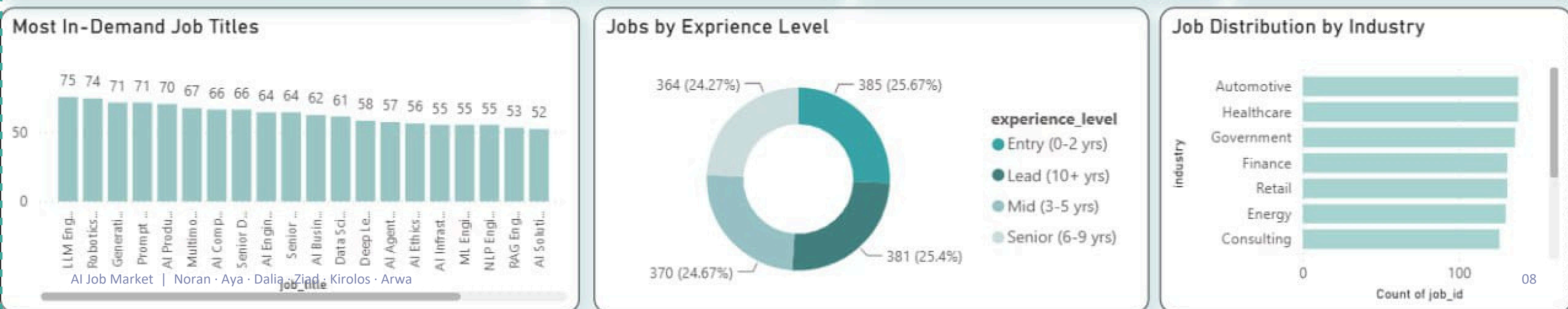
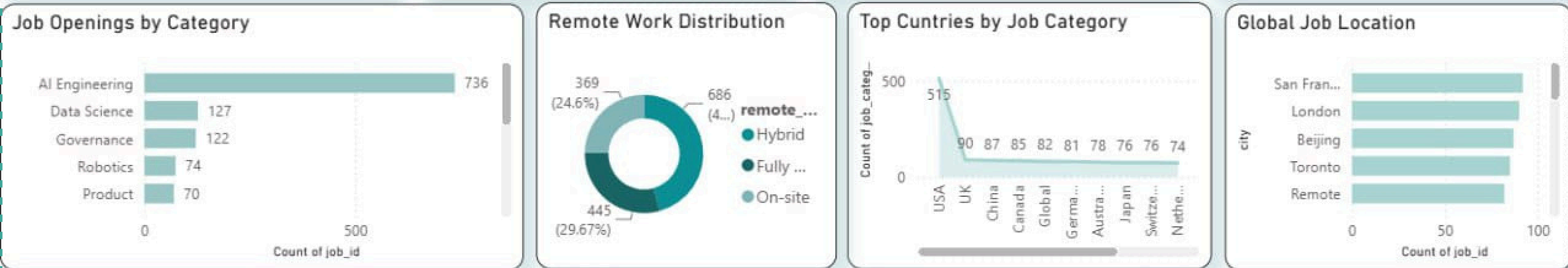


job_title
All

job_id
All

job_category
All

90,000.00384,000.00



Salary & Compensation Analysis

OBJECTIVE

Analyze salary patterns in the AI job market and understand how salary varies by job title, experience level, location, company size and work arrangement.

TARGET AUDIENCE

Job Seekers • HR Teams • Recruiters • Career Planners

Main KPIs: Average Salary • Minimum Salary • Maximum Salary • Average AI Salary Premium %

\$194,892

Average Salary

\$90,000

Minimum Salary

\$320,000

Maximum Salary

10.86%

Avg. AI Salary Premium

Top 5 Highest-Paying Job Titles (\$)



Average Salary by Experience Level (\$)





AI Market Salary Insight

Avg Salary
194.89K



Max Salary
320K



Min Salary
90K



Avg AI Premium
10.86



Salary Range (USD)

90,000.00 384,000.00



Job Title

All

Remote Work

All

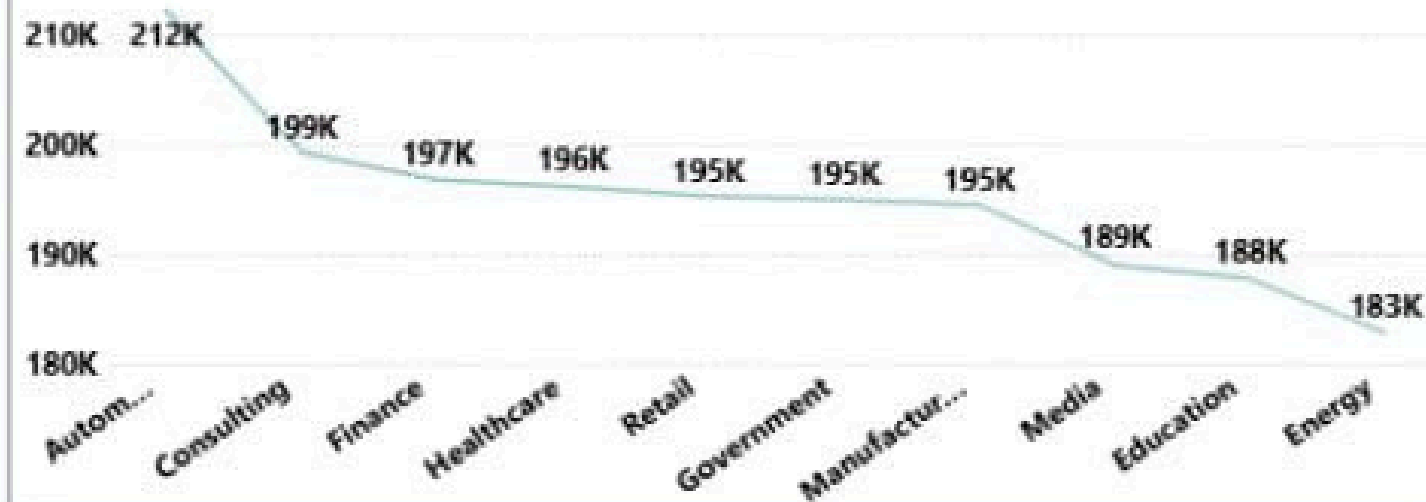
Industry

All

Experience Level

All

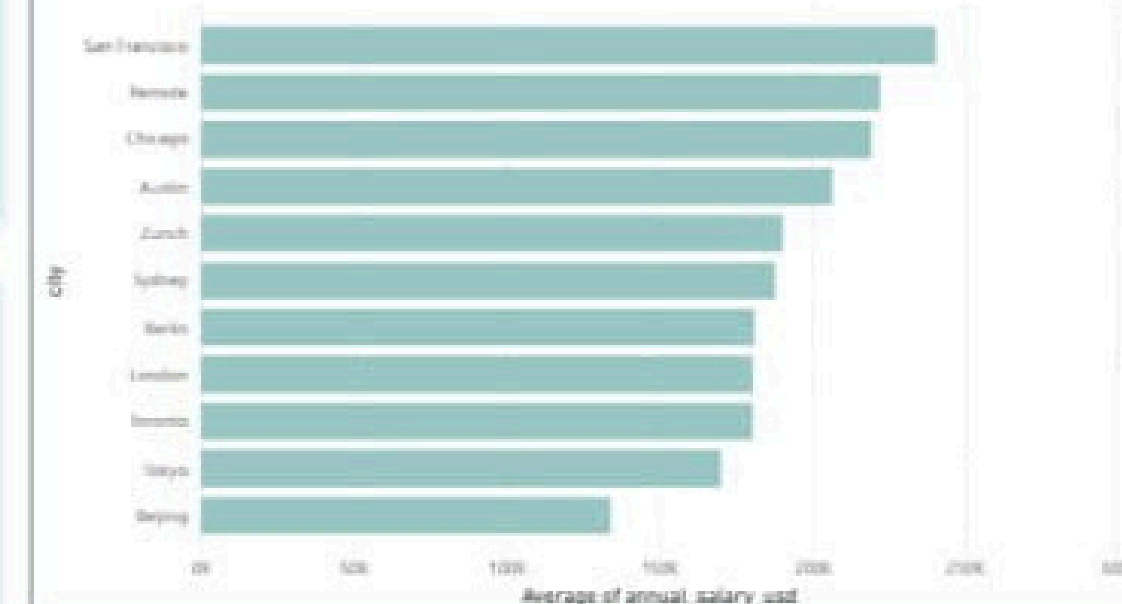
Average Salary By Industry



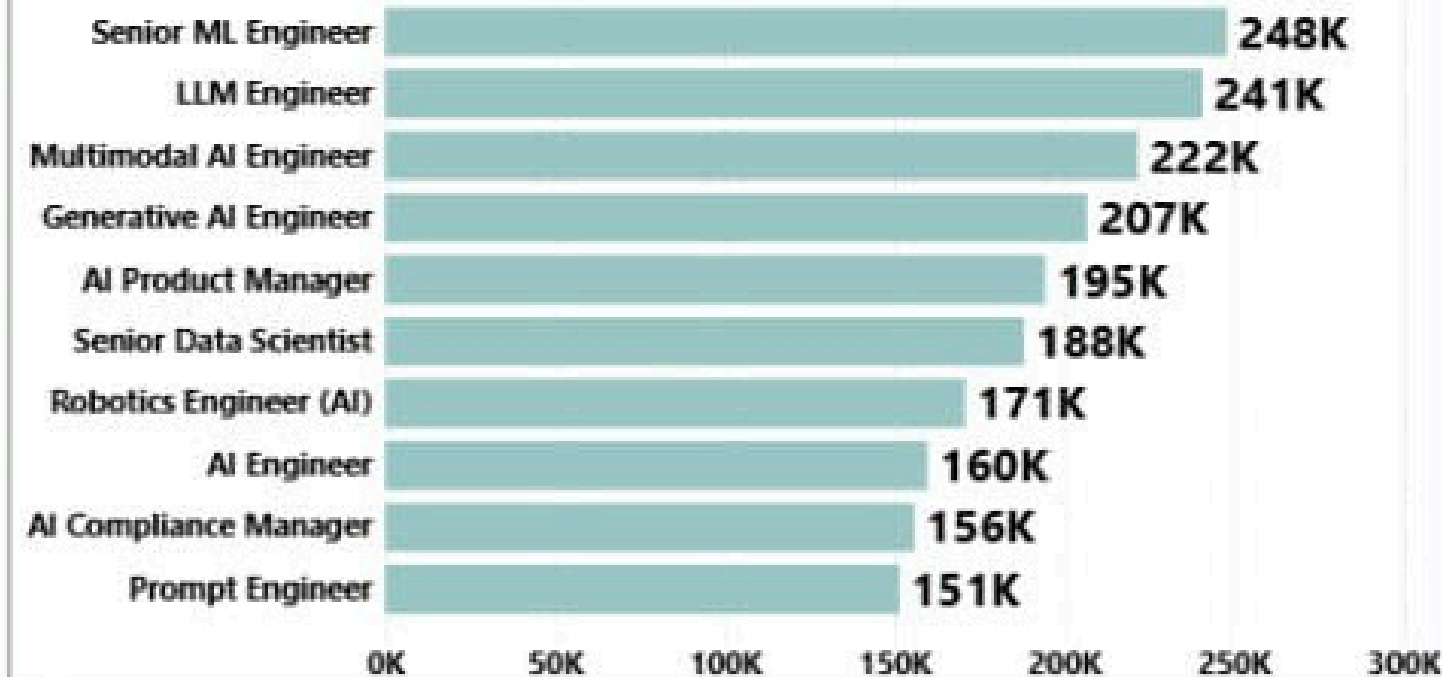
Average Salary By Work Arrangement



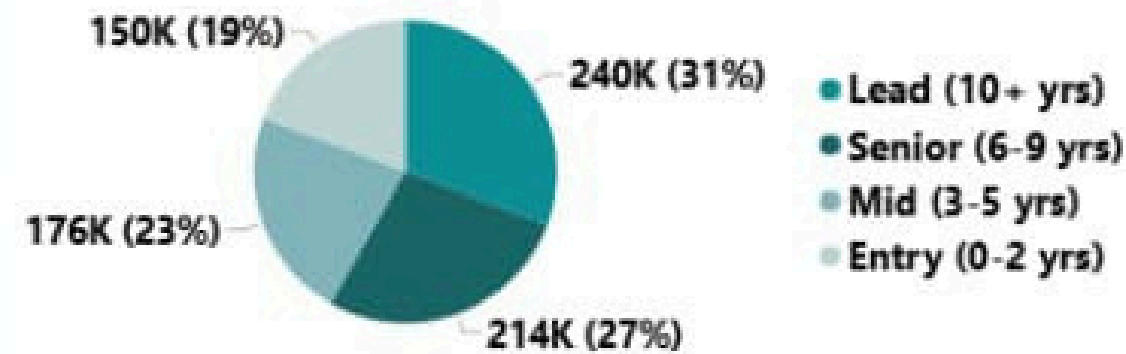
Average Salary By City



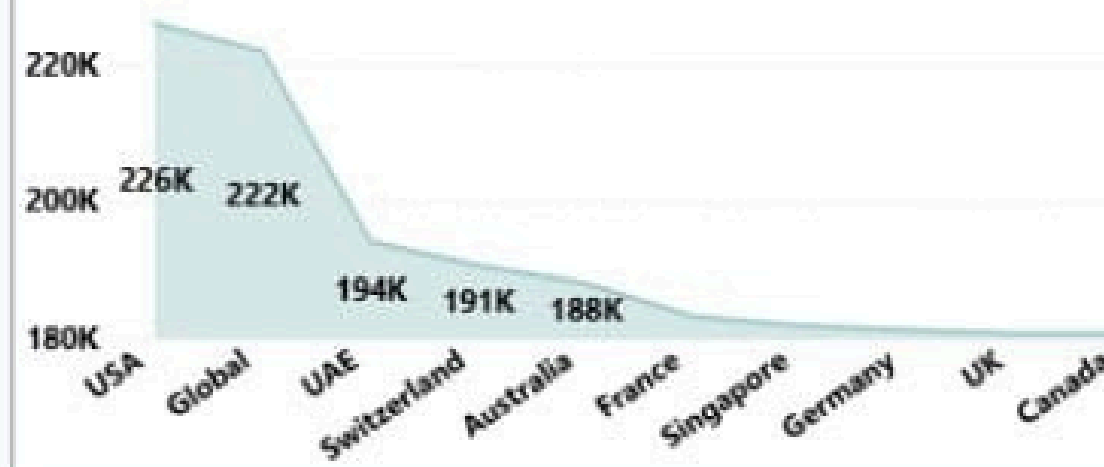
Average Salary By Job Title



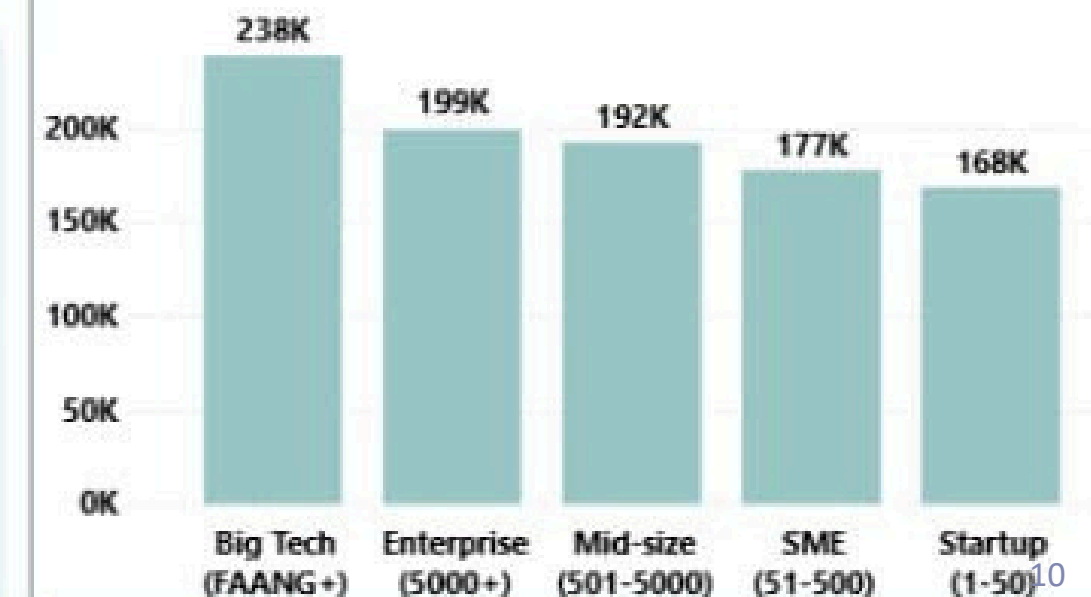
Average Salary By Experience Level



Average Salary By Country



Average Salary By Company Size



AI Skills & Career Requirements

OBJECTIVE

Identify the most demanded skills in the AI job market and understand how skill requirements vary across job roles, experience levels and salaries.

TARGET AUDIENCE

Job Seekers • Students • Career Planners • Recruiters

Main KPIs: Average Skills per Job · Maximum Skills Required · Most In-Demand Skill · % Jobs with Specified Skills

6.37

Average Skills per Job

10

Maximum Skills Required

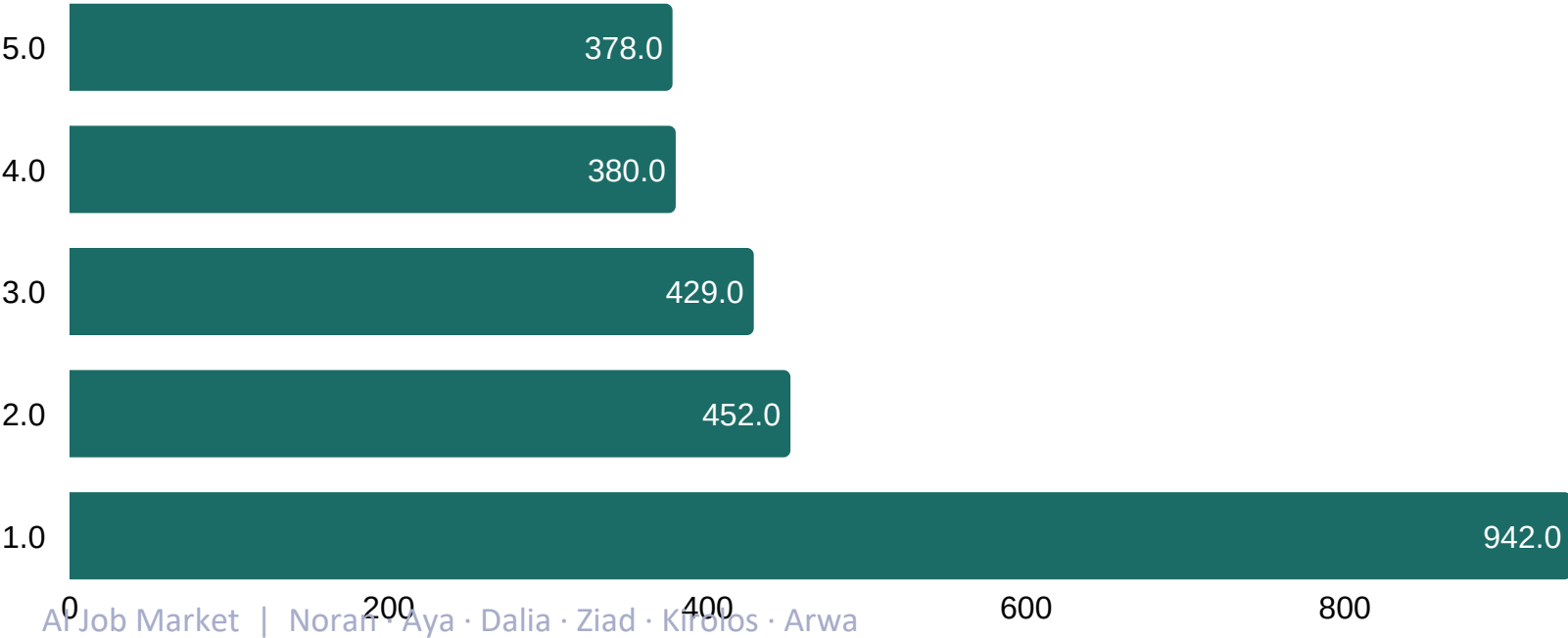
Python

Most In-Demand Skill

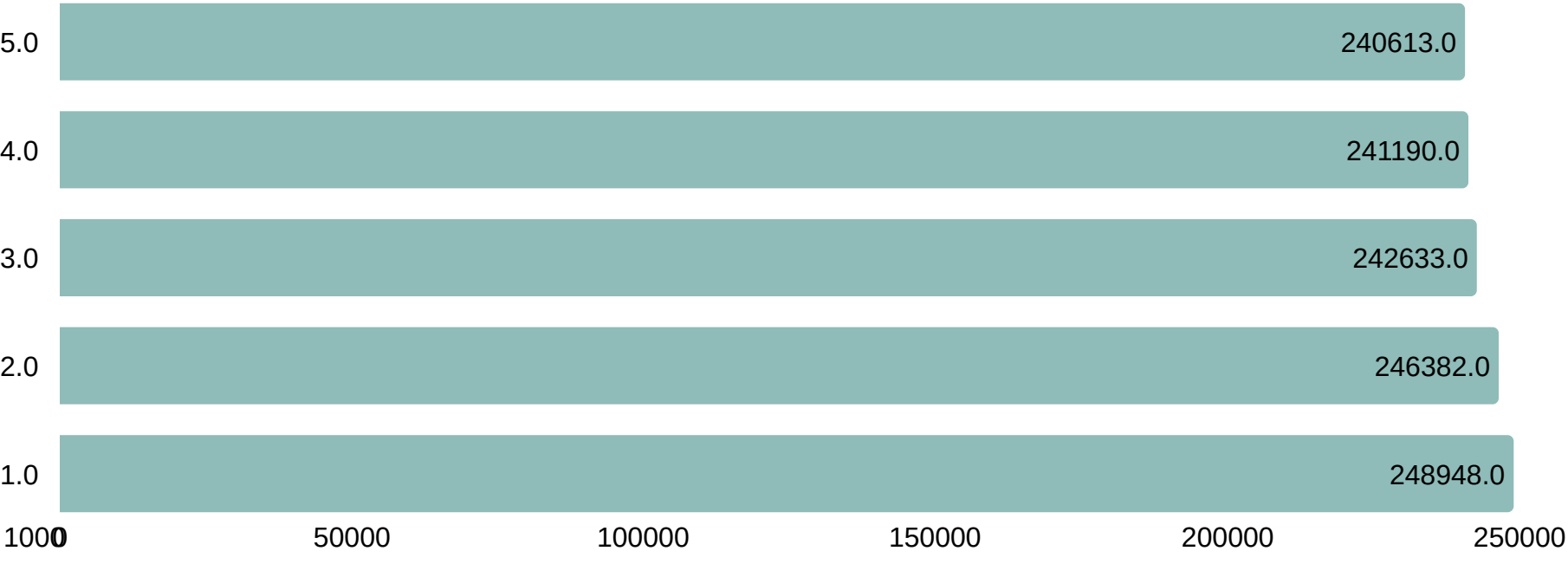
100%

Jobs with Specified Skills

Top 5 Most Required Skills (Postings)



Top 5 Skills at Highest-Paying Jobs (\$)





AI SKILLS & CAREER REQUIREMENTS

6.29



Average Skills Per Job

10



Maximum Skills Required

100%



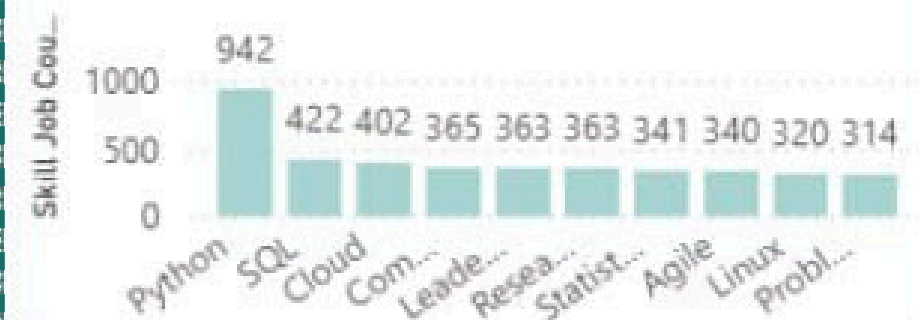
Percentage Jobs With Skills

Python

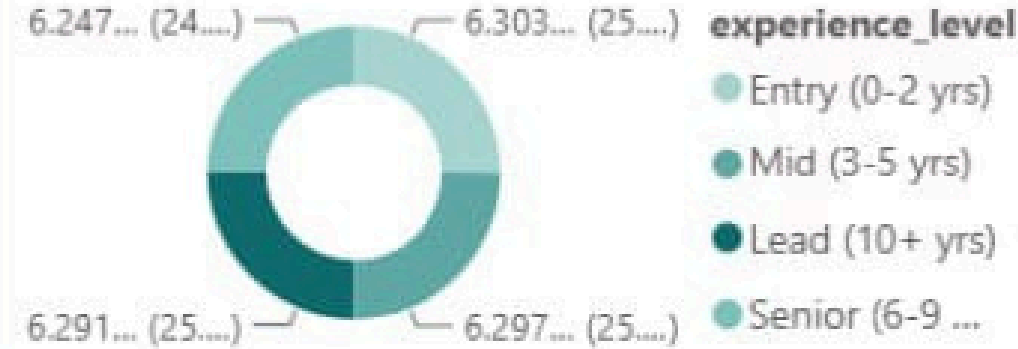


Most In-Demand Skill

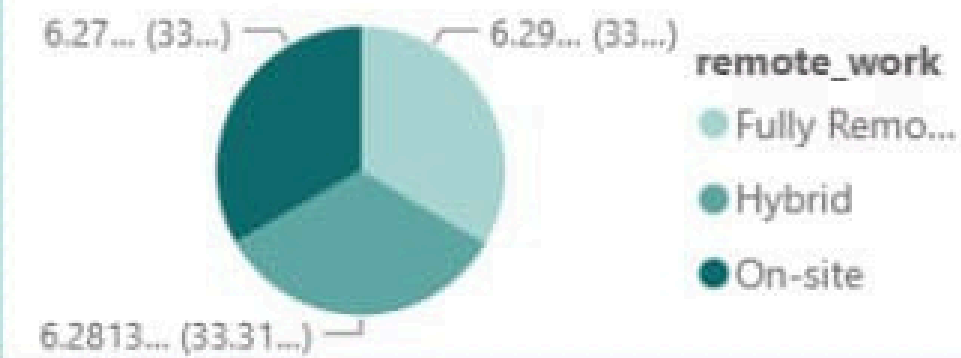
Top 10 Most In-Demand AI Skills



Average Skills Per Job by experience_level



Average Skills by Work Arrangement



experience_level

- ☐ Entry (0-2 yrs)
- ☐ Lead (10+ yrs)

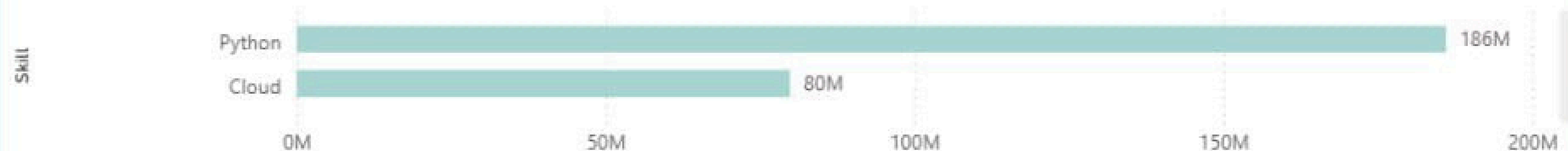
job_category

- ☐ AI Engineering
- ☐ Architecture

Top Skills Required for Senior Roles



Sum of annual_salary_usd by Skill



Average Skills Per Job by job_category



AI Job Demand & Market Trends

OBJECTIVE

Analyze AI job demand and market trends across job roles, job categories, industries, countries and experience levels.

TARGET AUDIENCE

Job Seekers • Career Analysts • Recruiters • Market Analysts

Main KPIs: Average Demand Score • Average Demand Growth % • Highest Demand Job Role • % LLM Roles

87.5 / 100

Average Demand Score

31.1%

Average Demand Growth

LLM Engineer

Highest Demand Role

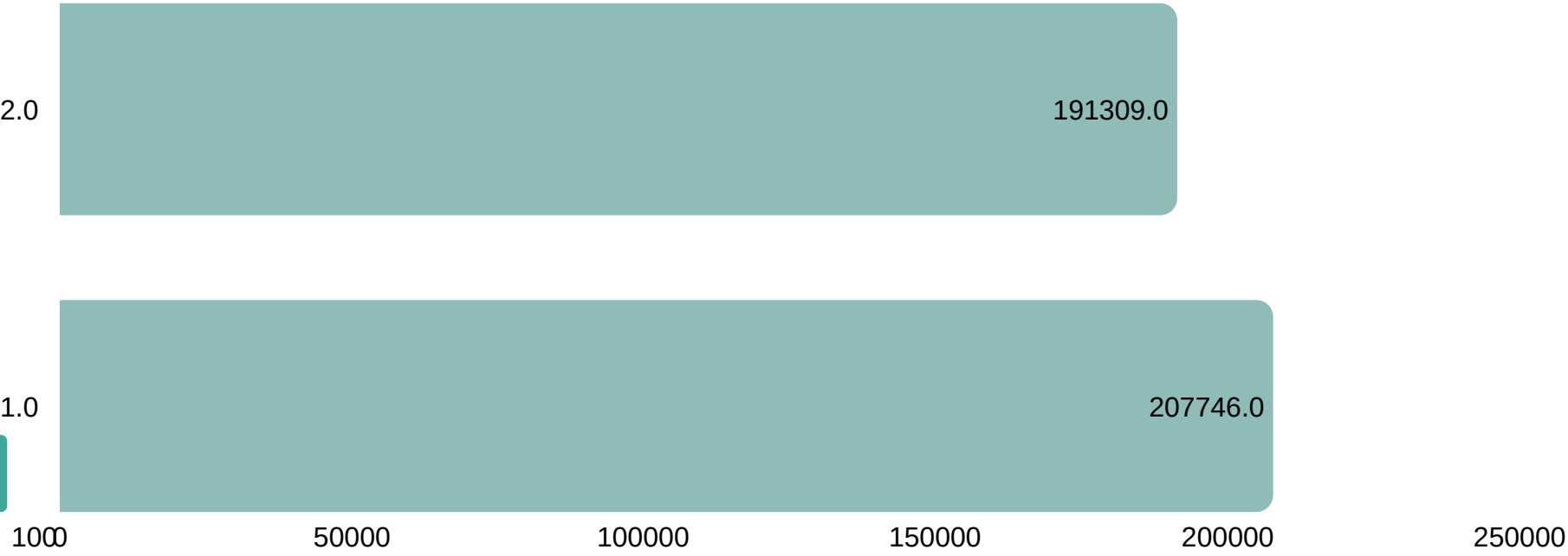
21.8%

LLM-Related Roles

Top 5 Job Titles by Demand Score



Avg. Salary: LLM vs Non-LLM Roles (\$)





AI JOB DEMAND & MARKET TRENDS

Average Demand Score
87.52

Average Demand Growth
31.12%

% LLM Roles
21.80%

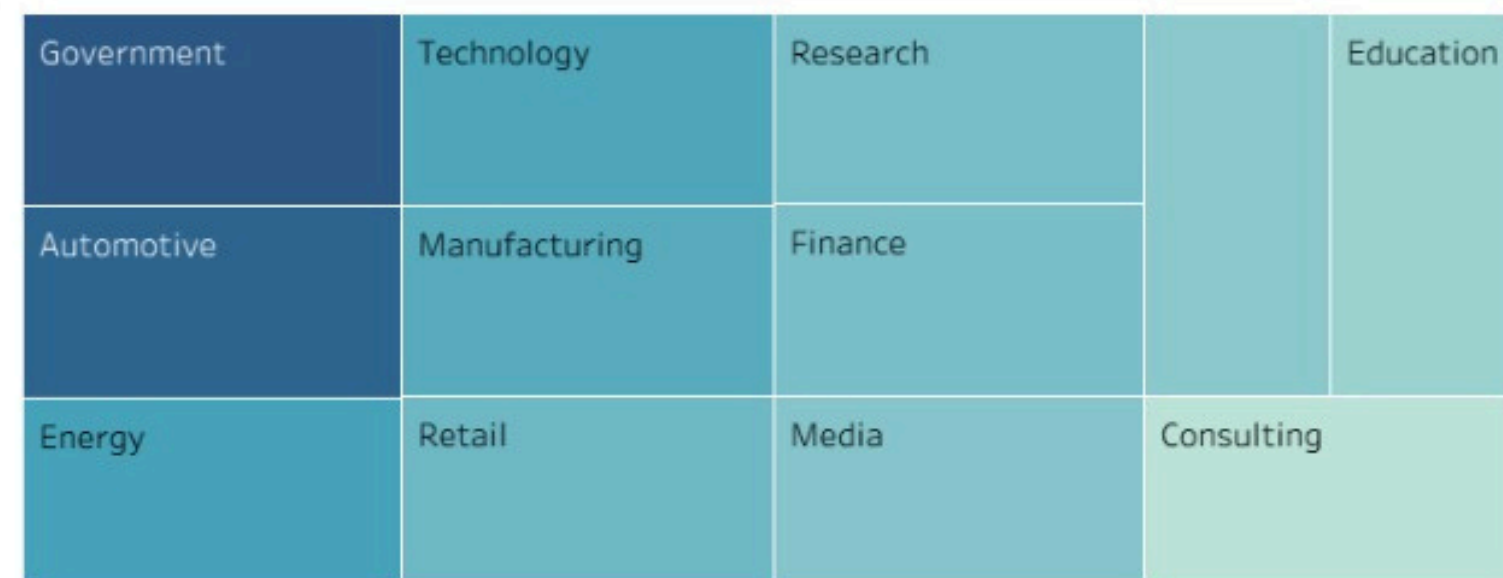
industry (All)

experien.. (All)

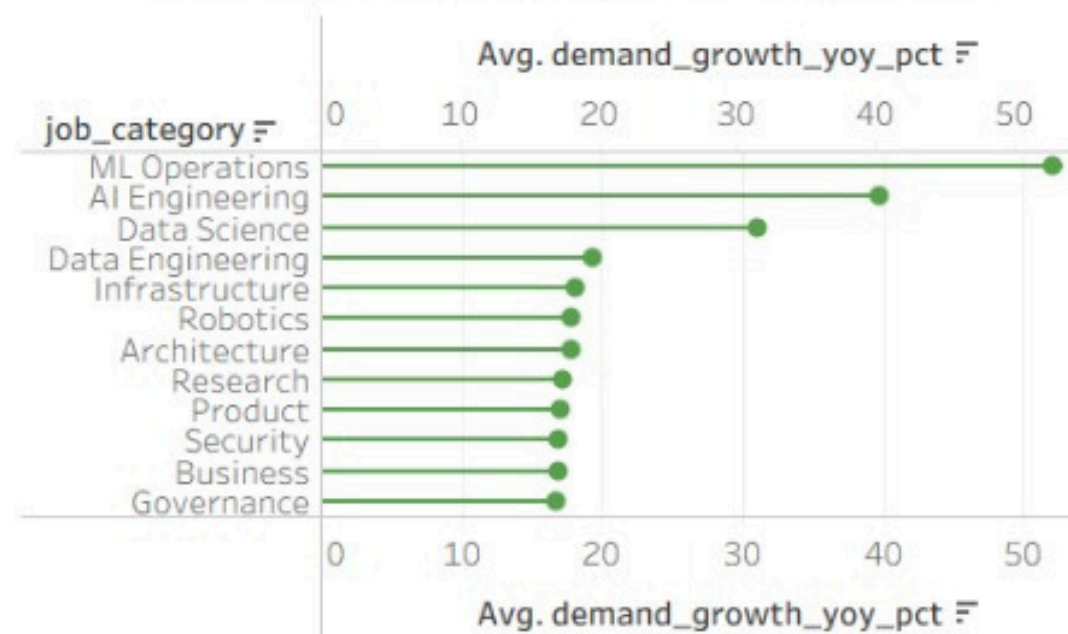
job_cate.. (All)

country (All)

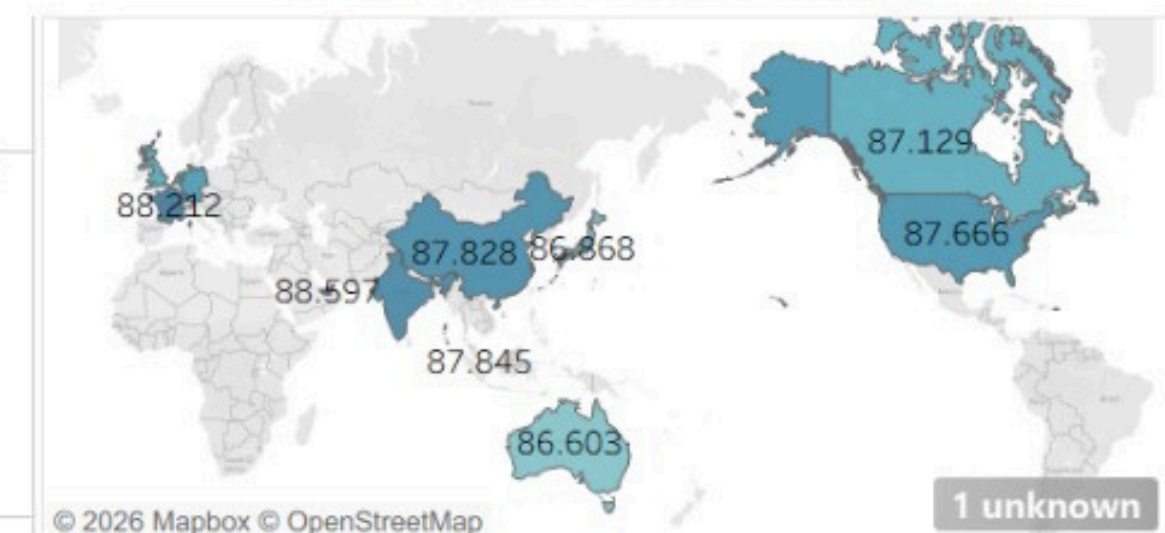
AI Demand by Industry



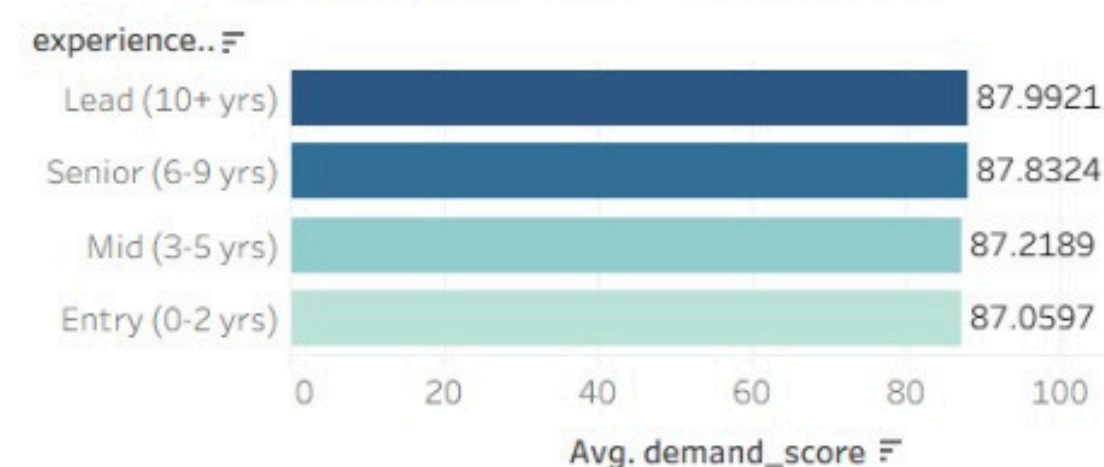
Demand Growth by Job Category



AI Job Demand by Country



Demand by Experience Level



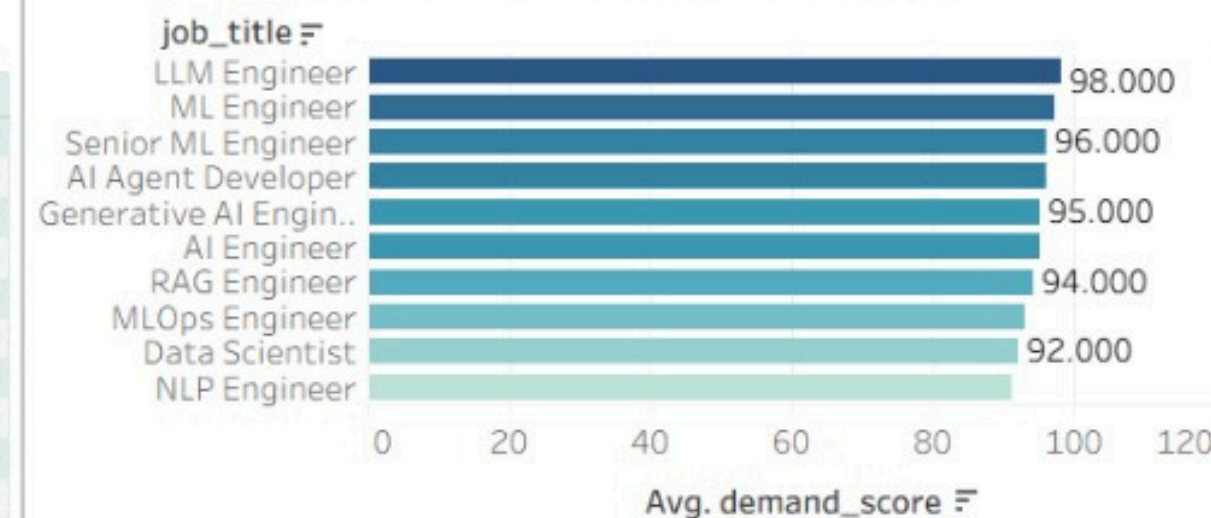
LLM vs Non-LLM Salary



Highest Demand Job Role

job_title	
LLM Engineer	98.000
ML Engineer	97.000
Senior ML Engineer	96.000
AI Agent Developer	96.000
Generative AI Engin..	95.000
AI Engineer	95.000
RAG Engineer	94.000
MLOps Engineer	93.000
Data Scientist	92.000
NLP Engineer	91.000

Top 10 AI Jobs by Demand





Key Insights From the Dashboards



Market Overview

The market is dominated by AI Engineering roles (736 of 1,500 postings) and led by the USA (515 postings), with Hybrid the most common work arrangement.



Compensation

Salary rises sharply with seniority — from \$150K at Entry level to \$240K at Lead level — and Big Tech (FAANG+) pays roughly 42% more than Startups.



Skills

Python appears in 942 of 1,500 postings (63%), yet the highest-paying skills are specialized: System Design, Prompt Engineering and RAG all average \$240K+.



Demand & Growth

LLM Engineer is both the highest-demand (98/100) and among the highest-paid roles; LLM-related roles pay ~\$16.4K more on average than non-LLM roles.

Business Conclusions

For Job Seekers

Prioritize LLM, RAG and system-design skills — they carry both the highest demand and the strongest salary premiums in the current AI job market.

For Recruiters

Competitive offers should benchmark against Big Tech pay bands (\$238K avg.); remote-friendly and hybrid roles now represent over 75% of postings.

For Career Planners

Skill depth matters more than skill count — average skills per job stay flat (~6.3) across experience levels, but which skills you hold drives salary.

For Market Analysts

LLM-related roles are the fastest-growing segment (demand score 98, +42.7% YoY for top titles) and command a clear salary premium over non-LLM roles.

Thank You

Questions & Discussion

AI Job Market — A Data-Driven Analysis

Noran Hatem • Aya Ayman • Dalia Khaled
Ziad Ashraf • Kirolos Ibrahim • Arwa Ahmed

Supervised by: Eng. Dina Mohsen

